Retail Analytics Project – Executive Summary Report

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Project: Month 1 – Retail Analytics (Tasks 1–4)

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1. Project Overview

This project focuses on building an end-to-end **Retail Analytics System** to analyze sales data, identify business insights, and develop predictive models for better decision-making.

The dataset used is a retail superstore dataset containing approximately **9,994 records** with features such as sales, profit, customer details, product categories, and order information.

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2. Data Preparation & Processing

- Cleaned dataset (no missing values, no duplicates)
- Converted date columns and extracted time-based features:
  - Year, Month, Quarter, Day of Week
- Created new features:
  - Profit Margin
  - Shipping Days
- Applied encoding techniques for categorical variables
- Scaled numerical features for machine learning
- Stored processed data in SQL database for structured access

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3. Exploratory Data Analysis (EDA)

Key Observations:

- **Technology category** generates the highest revenue
- **West region** contributes the highest sales
- Sales show **seasonal trends**, especially increasing toward year-end

Customer Insights:

- **Consumer segment** has the highest share
- **Corporate segment** has the highest Customer Lifetime Value (CLV)

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4. Feature Engineering

- Time-based feature creation (Month, Quarter, Year)
- One-hot encoding for categorical variables
- Interaction features for improving model performance
- Final dataset prepared with **38+ machine learning features**

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5. Model Development

Regression Models (Sales Prediction):

- Linear Regression
- Random Forest (Best Performing)
- XGBoost
- Ensemble (RF + XGBoost)

Time Series Models (Forecasting):

- ARIMA
- Prophet (Best Performing)

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6. Model Evaluation

Regression Performance:

- Random Forest RMSE ≈ **$500**
- Indicates average prediction error of ~$500 per order

Time Series Performance:

- Prophet MAPE ≈ **17–18%**
- Indicates ~82% forecasting accuracy

Key Finding:

- **Random Forest** is best for transaction-level prediction
- **Prophet** is best for time series forecasting

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7. Key Business Insights

Sales Drivers:

- **Quantity** is the most important factor affecting sales

- High-performing categories:
    - Machines
    - Copiers

Pricing Insights:

- Discounts above **30% lead to losses**
- Optimal discount range: **0%–20%**

Customer Insights:

- Corporate customers generate **highest lifetime value**
- Opportunity for targeted marketing

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8. Product & Market Insights

- Top products significantly contribute to revenue
- Sub-categories like **Phones and Chairs** dominate sales
- Regional performance shows **West and East leading markets**

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9. Forecasting Insights

- **ARIMA** provides stable but conservative forecasts
- **Prophet** captures trend and seasonality more effectively
- Prophet is recommended for **business planning and forecasting**

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10. Dashboard Overview

An interactive dashboard was developed including:

- Key KPIs (Sales, Profit, Orders, Customers)
- Monthly sales trends
- Customer segmentation
- Regional and category performance
- Sales drivers and product insights
- Pricing and discount analysis

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11. Business Recommendations

- Reduce high discounts (>30%) to prevent losses

- Focus on high-performing products (Machines, Copiers)
- Target high-value customers (Corporate segment)
- Use forecasting models for inventory and demand planning
- Optimize pricing strategy based on data insights

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12. Conclusion

This project successfully delivers a complete retail analytics solution, including:

- Data preprocessing and feature engineering
- In-depth exploratory analysis
- Machine learning and forecasting models
- Business insights and recommendations
- Interactive dashboard for decision-making

The solution is **industry-ready** and can support real-world business operations such as sales forecasting, pricing optimization, and customer targeting.

End of Report